

Riya Basak

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Profile

My research focuses on world models for general, reliable intelligence: learning structured representations of latent world state, discovering transferable dynamics, reasoning over imagined futures, and acting safely under uncertainty. I study self-supervised representation learning, compositional and causal world modelling, model-based decision-making and safe reinforcement learning, with applications across safety-critical scientific and embodied domains.

Technical Foundations

Programming & Research Computing: Python, C, Java, SQL, Bash, Linux, Git/GitHub, TeX/LaTeX

Mathematical Foundations: Linear Algebra, Probability, Statistics, Calculus, Optimisation

Machine Learning & Representation Learning: Self-Supervised Learning, Representation Learning, Multimodal Learning, CNN–Transformer Architectures, Explainable AI

World Models & Sequential Decision-Making: World Models, Model-Based Prediction, Offline Reinforcement Learning, Safe Reinforcement Learning, Constrained Policy Optimisation, Counterfactual Safety Analysis

Machine-Learning Frameworks: PyTorch, NumPy, scikit-learn, pandas, OpenCV, Matplotlib

Research Methodology: Controlled Baselines and Ablations, Leakage-Aware Validation, Out-of-Distribution Evaluation, Calibration, Predictive-Uncertainty Analysis, Failure-Mode Analysis, Reproducible Experimentation

Education

University of Hertfordshire

BSc (Hons) Computer Science with Artificial Intelligence — First Class Honours

Sep 2023 – May 2026

Hatfield, UK

GPA: Level 4: 4.34/4.50 | Level 5: 4.34/4.50 | Level 6: 4.44/4.50 | **Cumulative:** 4.38/4.50

Final-Year Departmental Standing: Top 1–3% in Computer Science | Top 1–2% in Artificial Intelligence

Experience

Cubble

Aug 2025 – Sep 2025

Full Stack AI Intern — ML Systems Research & Evaluation

- Investigated how deployed AI failures could be inspected systematically rather than through ad-hoc debugging; translated the proposed evaluation methodology into model-review, safety-flag, logging and failure-analysis tools.
- Integrated frontend inspection with backend AI services and structured testing to trace prediction errors, recurring failure modes and application-layer faults reproducibly.
- Improved the research-to-deployment workflow by making model behaviour and failure causes directly inspectable, with clearer experimental documentation and validation.

Cubble

Jul 2025 – Aug 2025

Software Engineer Intern — Machine Learning Research

- Investigated which learned visual representations generalised most reliably for safe/unsafe user-uploaded image classification under limited task-specific data; compared ResNet50 and CLIP-based approaches.
- Independently implemented transfer-learning pipelines and evaluated them through controlled validation, optimisation, quantitative error analysis and overfitting analysis to refine model design.
- Achieved **97.5%** accuracy on unseen images with stable validation behaviour; the AI team supervisor recognised my work as the top-performing implementation of the proposed working-model design.

Mercor

Sep 2025 – Nov 2025

Artificial Intelligence Researcher — Mathematical Reasoning

- Investigated failure modes in AI mathematical reasoning by formulating Olympiad-level problems as explicit assumptions, constraints, intermediate claims and proof obligations.
- Derived and evaluated solution paths using constructive reasoning, case analysis, edge-case testing and rigorous assumption checking, then assessed AI-generated proofs for validity, completeness and efficiency.
- Converted recurrent errors—including unsupported implications, omitted cases and circular reasoning—into structured evaluation criteria for more precise analysis of mathematical reasoning failures.

Projects

Mitigating Shortcut Learning in Brain Tumour MRI Classification

Oct 2025 – Mar 2026

Dissertation Project [PREPRINT] [CODE]

Shortcut Learning, Medical Imaging, Explainable AI

- Proposed Pathology-Focused Disentanglement and Guided Semantic Token Evolution (PFD-GSTE), with variants A and B, for shortcut-learning mitigation without segmentation masks.
- Designed a ResNet50V2–RvIT (CNN–Transformer) hybrid for four-class brain-tumour MRI classification; my literature review found no prior peer-reviewed study using this exact combination for the proposed shortcut learning mitigation experiment.
- Trained fixed-split PyTorch models with leakage-aware preprocessing and ablations; all models reached **98.52–99.22%** test accuracy and **98.49–99.20%** macro-F1.
- XAI analysis showed the main result: PFD-GSTE-B gave stronger tumour-region attribution than PFD-GSTE-A, with tumour-focused evidence in about **90%** versus **70%** of reviewed test cases.
- PFD-GSTE-B mitigated shortcut reliance more reliably on the tested benchmark; PFD-GSTE-A was weaker despite high classification accuracy.
- Also performed a small qualitative OOD demonstration on five external T1ce meningioma MRI images, kept separate from the formal benchmark evaluation and project submission.

Admissibility-Aware Offline Actor-Critic Learning for Safer ICU-Sepsis Treatment Decisions

Mar 2026 – Apr 2026

Proposed Coursework [CODE]

Safe Offline Deep RL

- Proposed Lagrangian Admissibility-Aware Deep Action-Nudging Actor-Critic (LAADAN-AC), an offline actor-critic method for safer treatment-policy learning in the ICU-Sepsis benchmark.
- Built, to my literature-review findings, a new combination for ICU-Sepsis: hard admissibility masking, twin critics, conservative critic regularisation, expert-policy regularisation, smoothness shaping, and Lagrangian cost control.
- Compared LAADAN-AC with Behaviour Cloning, CQL-regularised fitted-Q, and Vanilla Offline Actor-Critic under the same dataset, seeds, horizon, and evaluation metrics.
- LAADAN-AC achieved the strongest safe policy: **0.7931 ± 0.0007** survival/return, **0.0000** inadmissibility, and the highest expert-action match at **0.9525**.
- LAADAN-AC gave the best return–safety trade-off, improving over BC and CQL while avoiding the unsafe behaviour of unconstrained actor-critic learning.

LAADAN-AC: Beyond Survival in Admissible Offline Treatment-Policy Learning

Apr 2026 – Jun 2026

Under review [CODE]

Safe Offline Deep RL, Cross-Domain Check

- Extended LAADAN-AC into a reproducible research manuscript and codebase with component ablations, Lagrangian frontiers, safety-failure analysis, and a cross-domain portability check.

From World Models to Embodied Social Robots

May 2026 – Present

Research Project — in Preparation for TMLR

Embodied AI, Social Robotics, World Models

- Building a LeWM- and DreamerV3-based world-model stack for embodied agents that learn from predicted physical/social futures, using ARC Prize 2026 / ARC-AGI-3 as an initial abstract testbed before embodied AI and social robotics.
- Propose KCON: Kinetic Counterfactual Oversight Nexus, a compact executive layer between imagined futures and action choice, adding action-effect normlet memory, minimal intervention, uncertainty/novelty handling, and failure-aware selection.

Same Rules, New Worlds

Jul 2026 – Present

Research Project (Ongoing)

World Models, Causal Dynamics, Compositional Generalisation

- Investigating why world models that predict well in-distribution often fail to transfer learned dynamics across changes in context, appearance, embodiment and long-horizon interaction.
- Developing a mechanism-transport world model that separates persistent state from transferable dynamics, enforcing consistency across nuisance shifts while remaining sensitive to genuine changes in the underlying causal mechanism.
- Evaluating compositional and interventional generalisation under matched appearance–dynamics shifts, with controlled baselines, long-horizon tests and ablations before reporting quantitative claims.